

VIDHYA T R

AI Full Stack Engineer & Technical Product Manager | React | TypeScript | LLM Products

Germany (relocating — Chancenkarte work authorization, available immediately)

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PROFILE

AI Full Stack Engineer and Technical Product Manager with 5+ years of experience building enterprise web applications, applied AI workflows and cloud-based products across fintech, private credit, legal-tech and ESG domains. Strong hands-on experience with React, TypeScript, Python/FastAPI, NestJS, REST API design and integration, AWS, authentication, databases and production deployments. Experienced in LLM-based document workflows, RAG concepts, prompt engineering, AI agents, vector search and human-in-the-loop review systems. Takes products from zero to delivery — discovery, scoping and hands-on build. Work-authorized in Germany via Chancenkarte and available immediately.

CORE TECHNICAL SKILLS

Frontend: React.js, TypeScript, JavaScript, Next.js, Vite, HTML, CSS, responsive UI, AG Grid, MUI, PSPDFKit, microfrontends

Backend and APIs: Python, FastAPI, NestJS, Node.js, Django, Flask, ASP.NET MVC, REST API design, API integration, authentication, RBAC, OAuth, SSO

AI / LLM Engineering: LLM workflows, RAG, prompt engineering, AI agents, agentic workflows, embeddings, vector search, document extraction, taxonomy design, model/provider orchestration, AI evaluation, human-in-the-loop review

Cloud and DevOps: AWS S3, Lambda, ECS Fargate, API Gateway, DynamoDB Streams, OpenSearch, Cognito, CloudWatch, Glue, Athena, CDK, Docker, Nginx, Gunicorn, CloudFront

Databases, Search and Testing: PostgreSQL, MSSQL, MongoDB, DynamoDB, Redis, OpenSearch/Elasticsearch, Jest, Playwright, unit testing, E2E testing

Engineering Practices: Git, code reviews, module federation, single-spa, shared libraries, CI/CD mindset, Agile/Scrum, product engineering

PROFESSIONAL EXPERIENCE

Technical Product Manager and Hands-on Full Stack Engineer - Alphastream.ai | Mar 2025 - Apr 2026 | India

- Led delivery of Document Intelligence, a production AI-powered product for document upload, taxonomy creation, extraction and review workflows, while staying hands-on across React/TypeScript UI and backend integration work.
- Worked as an applied AI product engineer close to stakeholders, translating client needs into user stories, acceptance criteria, release-ready features and customer-facing demos.
- Built and refined template-driven IC Memo and IDR Memo workflows, connecting frontend journeys with retrieval-based backend services, prompt-driven extraction and review flows.
- Reduced manual document-processing effort by about 80%, lowered operational cost by more than 50% and improved client satisfaction by about 15%.

Full Stack Developer - Alphastream.ai | Jan 2024 - Feb 2025 | India

- Developed scalable enterprise web applications using React, TypeScript, Next.js, NestJS APIs, Python/FastAPI components and AWS services for 5+ enterprise clients.
- Designed and integrated REST APIs for authentication, document workflows, search, review and client-facing product features.
- Worked with PostgreSQL, MongoDB, MSSQL and DynamoDB across application data, audit history, search workflows and analytics pipelines.
- Added unit and E2E test coverage using Jest and Playwright for critical user workflows, improving release confidence and regression testing.
- Moved a legacy monolithic frontend toward microfrontends using single-spa and module federation, improving modularity, ownership and deployment flexibility.
- Built multi-tenant authentication with Okta and Azure AD, supporting RBAC, secure access control and GDPR-aligned client tenant separation.
- Contributed to event-driven search and analytics pipelines using DynamoDB Streams, AWS Lambda, OpenSearch, S3, Glue and Athena.
- Provisioned and supported AWS infrastructure with CDK, VPC, ECS Fargate, API Gateway, Cognito and CloudWatch for production workloads.

Software Developer - Alphastream.ai | Jul 2022 - Dec 2023 | India

- Built React/TypeScript applications used by financial analysts to process, annotate, compare and extract clauses from complex legal and credit documents.
- Integrated AG Grid, PSPDFKit and MUI to support high-performance data grids, PDF analysis, AI-assisted review and business-user workflows.
- Designed calculation and validation features for interest schedules, covenant checks, dynamic formulas and business logic.
- Developed Django APIs, secure REST services, role-based access control, audit history and OpenSearch/Elasticsearch-backed search features.

Software Engineer - Tata Consultancy Services - H&M Client | Jan 2021 - Jul 2022 | India

- Maintained and improved systems supporting H&M tag design workflows, with exposure to ASP.NET MVC, APIs, SQL and Azure-based solutions.
- Optimized SQL queries and stored procedures, improving execution time, throughput and database reliability; built data integration pipelines using Azure Data Factory and SQL.

PERSONAL PROJECTS

- **Agentry - Agentic Content Automation Engine:** Built a config-driven, self-hosted AI workflow automation engine with Python, JSON workflows, provider/plugin registration, visual builder concepts, GitHub Actions deployment, cost-aware LLM orchestration, safety guardrails and dashboard monitoring.
- **MyWorkSpace - Universal AI Agent and Agentic Office Execution System:** Built a FastAPI-based AI workspace where a founder provides product requirements and a routing brain assigns work to specialist agents including PM, design, QA and delivery agents. Implemented skill registry, session memory, SQLite logs, fallback handling, mock-tested LLM integration and a React/TypeScript skill configuration panel.
- **TELC Exam Preparation App:** Built a German exam preparation web app for TELC/A1 learning with structured practice, vocabulary, grammar support, mock-style exercises and progress-oriented study flows for Germany-focused career preparation.

EDUCATION

B.Tech, Computer Science and Engineering - Government Engineering College Sreekrishnapuram, Kerala, India | 2016 - 2020 | CGPA: 8.9/10

CERTIFICATIONS AND LANGUAGES

Certifications: Certified ScrumMaster (CSM), Scrum Alliance | Microsoft Certified: Azure Fundamentals (AZ-900)

Languages: English - Professional working proficiency; German - A1 beginner, actively learning; Malayalam - Native; Hindi - Intermediate; Tamil - Conversational